mseg_counts/nEIL

MC vs data · overview · rank 80 / 98

rank 80 / 98 by endcap D_equal

used by: neither NN list

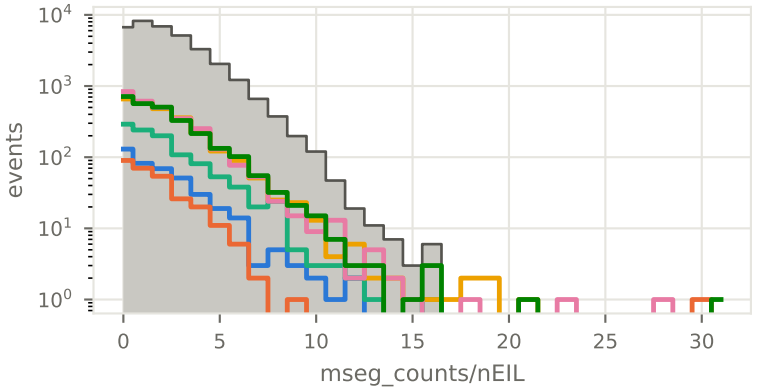
region	D_equal	noise	D_pooled	125/15	125/55	200/50	400/100	600/150	1000/275	n_bkg	rank
endcap	0.093	0.022	0.080	0.125	0.142	0.080	0.057	0.098	0.072	34,970	80/98
barrel	0.170	0.051	0.149	0.205	0.201	0.190	0.150	0.141	0.133	1,089	58/98
all	0.181	0.018	0.169	0.202	0.220	0.171	0.144	0.186	0.165	36,059	75/98

mseg_counts/nEIL

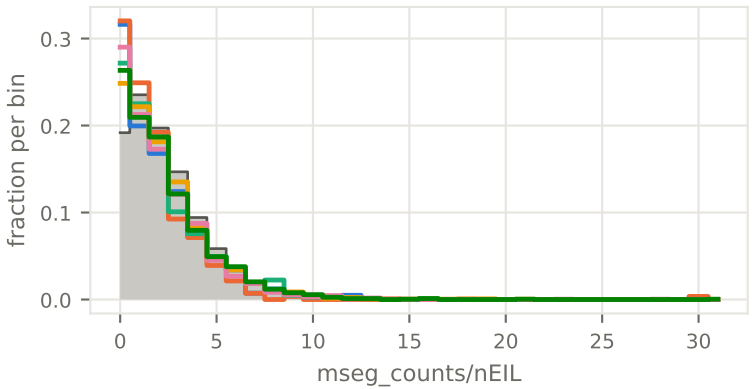
region: ENDCAP · rank 80 / 98

D_equal 0.093 (noise 0.022) D_pooled 0.080 min 0.057 (mH400_mS100) max 0.142 (mH125_mS55) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

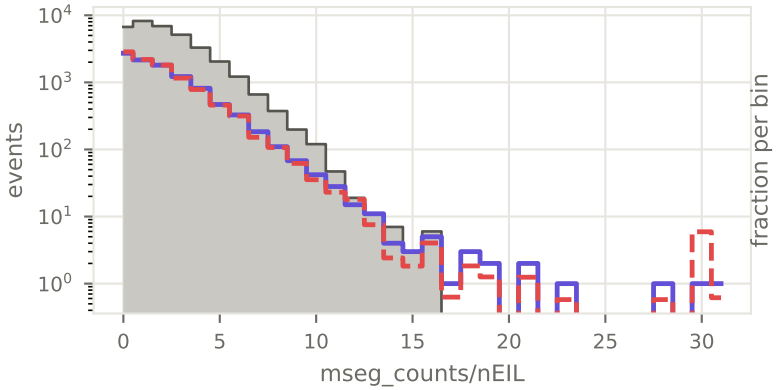


1b · each mass point vs background · normalized

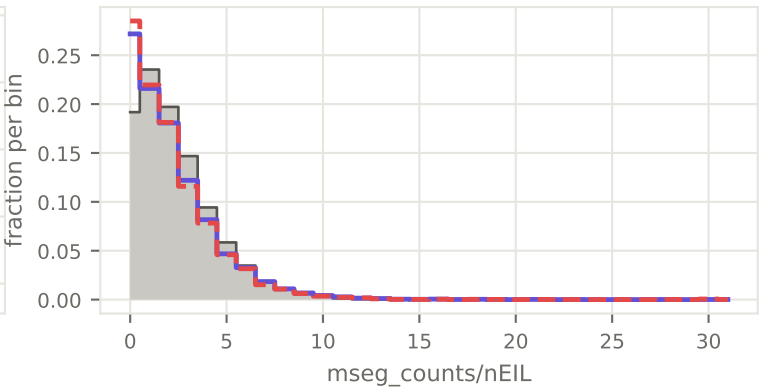


1 · each mass point
- background n 34,970
- mH125_mS15 D 0.125 n 411
- mH125_mS55 D 0.142 n 281
- mH200_mS50 D 0.080 n 1,071
- mH400_mS100 D 0.057 n 2,648
- mH600_mS150 D 0.098 n 2,882
- mH1000_mS275 D 0.072 n 2,703

2a · combined signal vs background · raw

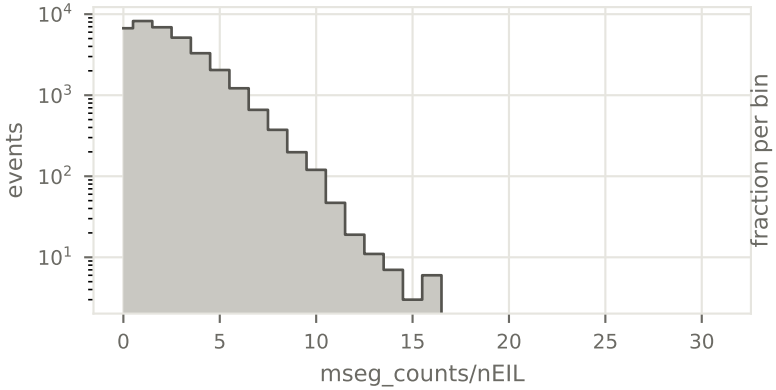


2b · combined signal vs background · normalized

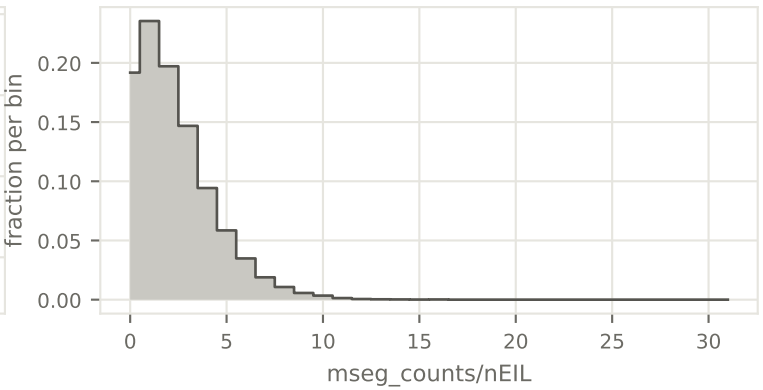


2 · combined signal
- background n 34,970
- combined, pooled D 0.080 n 9,996
- equal weight per mass D 0.093

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
- background n 34,970

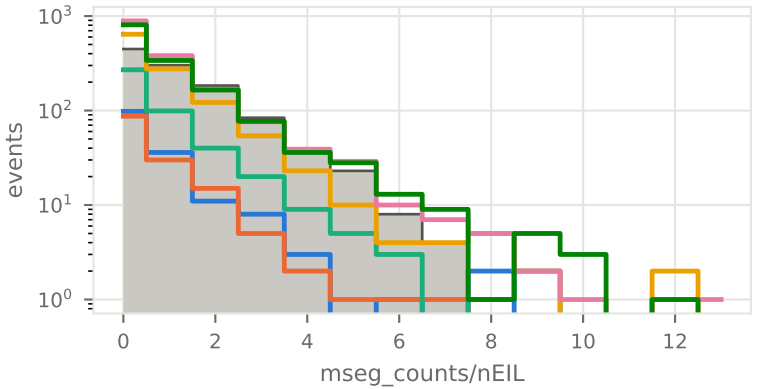
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

mseg_counts/nEIL

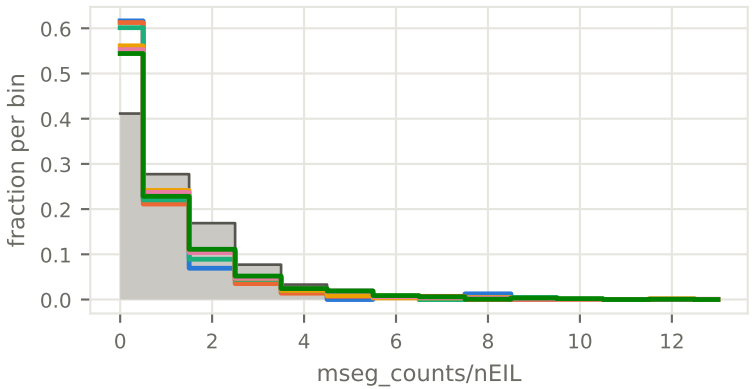
region: BARREL · rank 80 / 98

D_equal 0.170 (noise 0.051) D_pooled 0.149 min 0.133 (mH1000_mS275) max 0.205 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

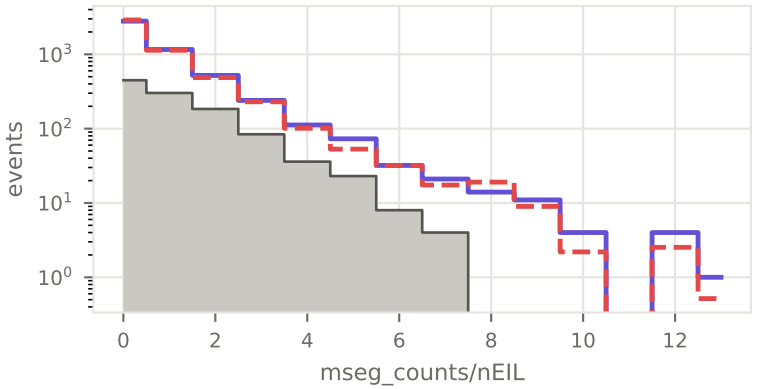


1b · each mass point vs background · normalized

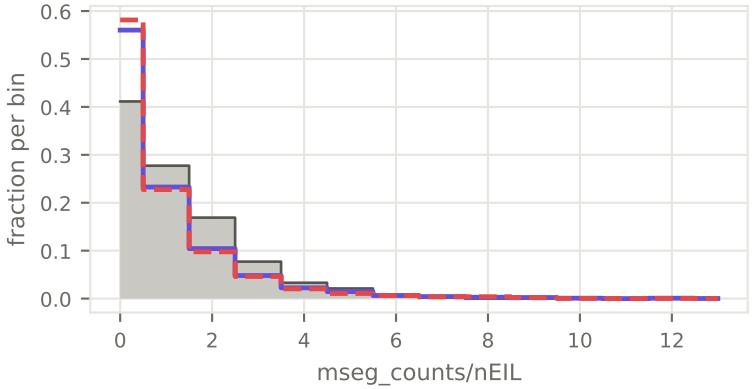


1 · each mass point
background n 1,089
mH125_mS15 D 0.205 n 159
mH125_mS55 D 0.201 n 142
mH200_mS50 D 0.190 n 449
mH400_mS100 D 0.150 n 1,144
mH600_mS150 D 0.141 n 1,608
mH1000_mS275 D 0.133 n 1,485

2a · combined signal vs background · raw

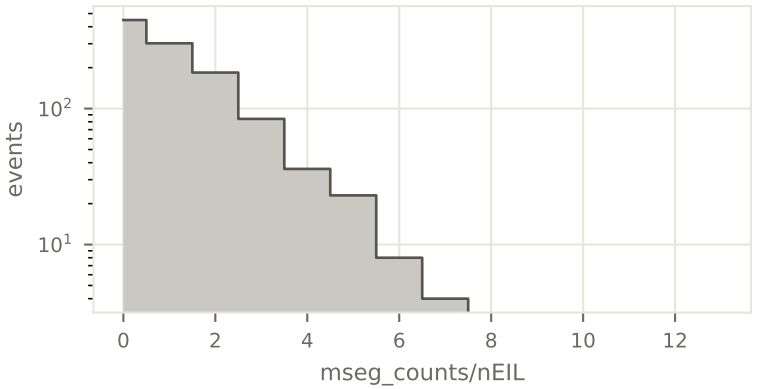


2b · combined signal vs background · normalized

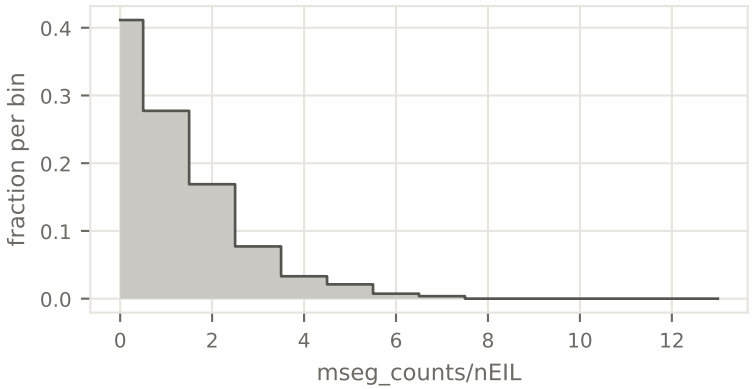


2 · combined signal
background n 1,089
combined, pooled D 0.149 n 4,987
equal weight per mass D 0.170

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

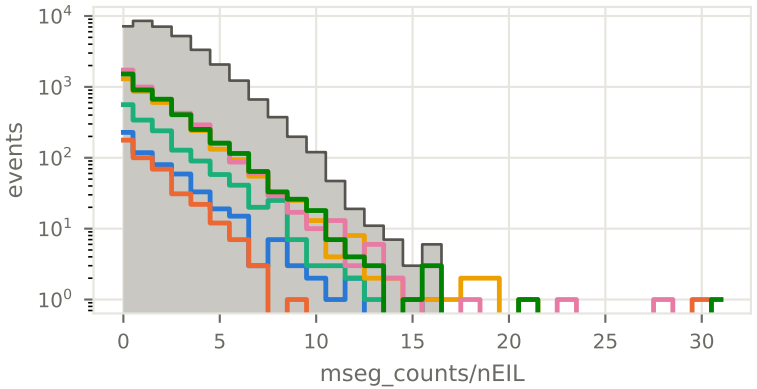
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

mseg_counts/nEIL

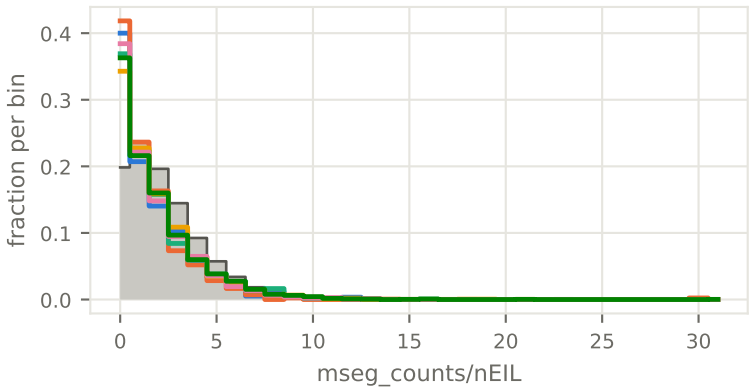
region: ALL · rank 80 / 98

D_equal 0.181 (noise 0.018) D_pooled 0.169 min 0.144 (mH400_mS100) max 0.220 (mH125_mS55) bkg n = 36,059 sig n = 14,983

1a · each mass point vs background · raw

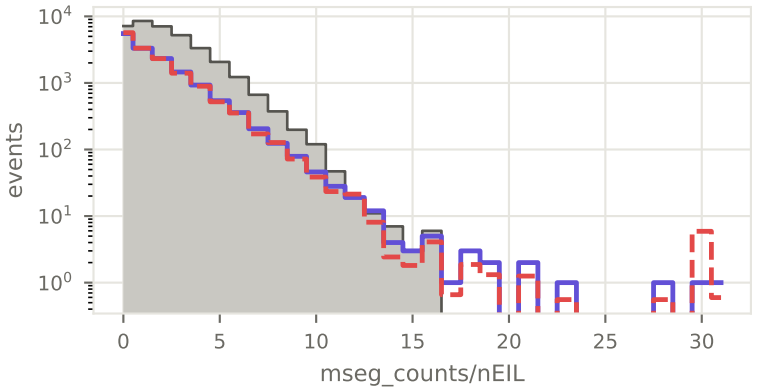


1b · each mass point vs background · normalized

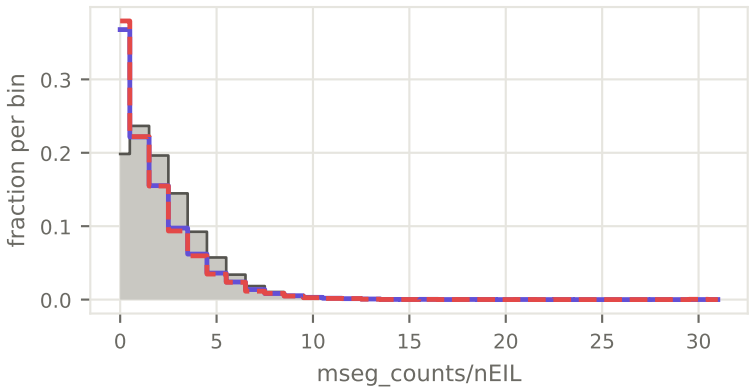


1 · each mass point
background n 36,059
mH125_mS15 D 0.202 n 570
mH125_mS55 D 0.220 n 423
mH200_mS50 D 0.171 n 1,520
mH400_mS100 D 0.144 n 3,792
mH600_mS150 D 0.186 n 4,490
mH1000_mS275 D 0.165 n 4,188

2a · combined signal vs background · raw

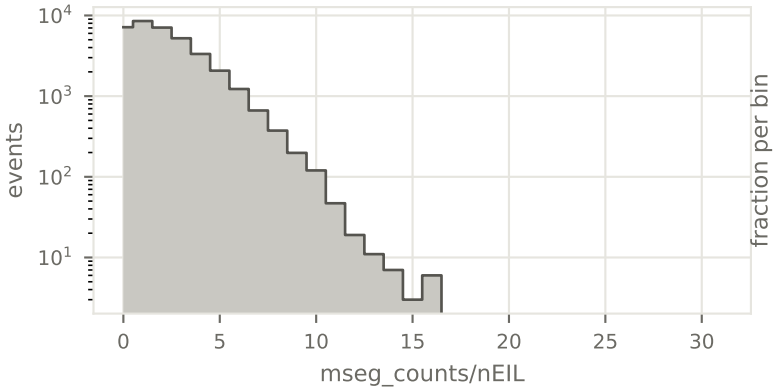


2b · combined signal vs background · normalized

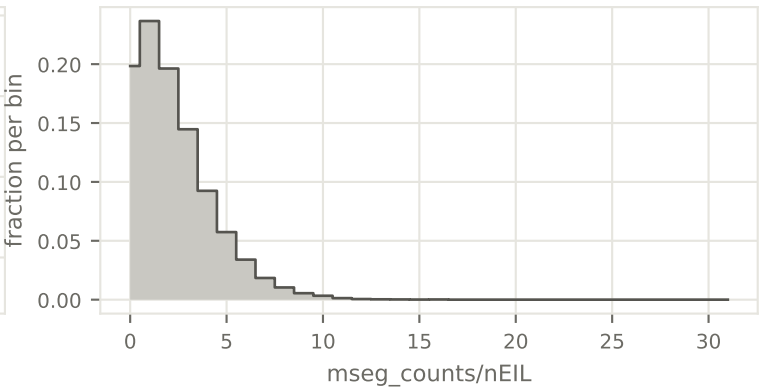


2 · combined signal
background n 36,059
combined, pooled D 0.169 n 14,983
equal weight per mass D 0.181

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 36,059

a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed